

EVENT MANAGEMENT SYSTEM

Functional, Non-Functional Requirements and Requirement Analysis

1. Functional and Non-Functional Requirements

1.1 Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale / Comments
FR-01	Functional	New users choose to register as an Organizer or an Attendee, and that choice sets which dashboard and tools they get from then on.	High	Pass: an Organizer sees event-creation tools; an Attendee sees their tickets and browsing options. Fail: an Attendee reaches the event-creation screen or views another user's ticket history.	Keeps host and attendee workflows and data fully separate.
FR-02	Functional	Organizers create an event with title, date, time, venue, ticket capacity, and price, then edit and publish it.	High	Pass: publishing makes the event show up on public listings right away. Fail: it publishes without a capacity limit, or doesn't appear publicly.	Everything else depends on an event existing to sell tickets against.
FR-03	Functional	Attendees browse active events, filter by date or category, and open full event details.	Medium	Pass: searches return accurate, active events in under 2 seconds. Fail: past or canceled events still show up by default.	Easy discovery drives both sales and attendance.
FR-04	Functional	Starting checkout puts the requested ticket quantity on a temporary hold; if there aren't enough left, checkout stops right away.	High	Pass: held tickets become unavailable to others immediately; insufficient stock blocks checkout before payment. Fail: two buyers can hold the same last ticket, or a shortfall only shows up at payment.	A one-time availability check isn't enough when two people check out at the same moment.
FR-05	Functional	Before payment, the system validates the Attendee's contact details and blocks checkout on missing fields or a bad email.	Medium	Pass: a blank field or invalid email is flagged and stops the order. Fail: a clearly broken email passes through to payment.	A ticket is useless if the receipt can't reach the buyer.
FR-06	Functional	An approved payment converts the hold into a permanent deduction, generates a ticket, and sends a confirmation and receipt; a decline releases the hold with a retry/cancel option.	High	Pass: approval produces a Ticket ID and confirmation; a decline releases the hold with a clear error. Fail: a decline still consumes tickets, or an approval produces no ticket.	Payment, inventory, and ticket issuance must always stay in sync.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale / Comments
FR-07	Functional	Attendees can cancel before the event, with refunds following the Organizer's policy.	Medium	Pass: canceling voids the ticket and frees the seat. Fail: a canceled ticket still counts as occupied.	Lets Organizers resell freed seats instead of losing them.
FR-08	Functional	Organizers can download a manifest of confirmed attendees with names, emails, and ticket IDs.	Medium	Pass: the CSV includes only paid, confirmed attendees. Fail: canceled tickets appear as active.	Needed for check-in and post-event follow-up.
FR-09	Functional	Platform Admins can suspend events or Organizer accounts that violate terms of service.	High	Pass: suspension removes public listings and blocks new sales instantly. Fail: a suspended event can still be bought via direct link.	The platform's main defense against fraud and policy violations.

1.2 Non-Functional Requirements

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale / Comments
NFR-01	Non-Functional (Performance)	Checkout must handle up to 10,000 concurrent buyers with pages loading in under 3 seconds.	High	Pass: load tests confirm sub-3-second responses during a simulated flash sale. Fail: the system slows or times out under peak traffic.	Traffic spikes during popular sales; a slow checkout costs money and trust.
NFR-02	Non-Functional (Security)	Full card numbers are never stored; all payment data is tokenized for PCI-DSS compliance.	High	Pass: audits find only tokens and masked card digits. Fail: a raw card number appears anywhere in storage or logs.	Avoids liability and keeps customer data safe.
NFR-03	Non-Functional (Reliability)	Ticket bookings use ACID-compliant transactions so two buyers can't claim the same last seat.	High	Pass: simultaneous purchases of the last ticket yield one success and one clear failure. Fail: both buyers get charged for one ticket.	Prevents double-booking and the refund cleanup it causes.
NFR-04	Non-Functional (Usability)	The Attendee experience is mobile-first, so checkout works smoothly on a phone.	Medium	Pass: layout adapts with no horizontal scrolling on mobile. Fail: buttons are too small or the payment form breaks on mobile.	Most tickets are bought from phones.
NFR-05	Non-Functional (Maintainability)	Payment processing stays loosely coupled from inventory logic via a defined API layer.	Low	Pass: swapping payment providers only touches the payment layer. Fail: payment logic is hardcoded into inventory queries.	Keeps vendor changes from requiring a system rewrite.

Requirement Analysis

1. User Requirements

The Event Management System shall provide users with the ability to:

- Register as either an Event Organizer or an Attendee, with role-based access to the matching dashboard.
- Create, edit, and publish events by specifying title, date, time, venue, ticket capacity, and price.
- Search and filter active events by date or category, and view full event details.
- Select a ticket quantity at checkout and be told immediately if there isn't enough availability.
- Have contact details validated before payment, so ticket and receipt delivery isn't put at risk.
- Pay securely by credit card and receive a digital ticket along with an email receipt.
- Have held tickets released back automatically if a payment attempt fails, so another attempt can be made.
- Cancel a registration before the event date, subject to the Organizer's refund policy.
- Export the list of confirmed attendees, including names, emails, and ticket IDs, for check-in and follow-up.
- Suspend events or Organizer accounts found to be fraudulent or in violation of platform terms.

2. System Requirements

The Event Management System shall:

- Provide a web-based platform with a centralized database for managing events, ticketing, and user accounts.
- Enforce session validation and role-based access control, routing users to the correct dashboard (Admin, Organizer, or Attendee).
- Place a temporary hold on requested tickets the instant checkout starts, using database-level locking to prevent double-selling.
- Release a ticket hold automatically back into available inventory if payment fails, and convert it to a permanent deduction once payment is approved.
- Validate contact-detail inputs, including required fields and email format, before an order reaches the payment step.
- Integrate with an external payment gateway over RESTful APIs to handle authorization, capture, and tokenization of payments.
- Expose a webhook endpoint to process payment gateway callbacks and trigger ticket generation or hold release accordingly.
- Integrate with a transactional email service to dispatch digital tickets and receipts without blocking checkout.
- Support exporting attendee and ticket data as a downloadable CSV manifest for Organizers.
- Provide fast response times, quick page loading, and reliable transaction processing under concurrent user load.
- Maintain data consistency, reliability, and integrity across bookings, cancellations, and payments.
- Provide secure, tokenized payment handling and encrypted data transfer.
- Support access across desktop, tablet, and mobile devices.

3. Domain Requirements

The Event Management System shall comply with requirements specific to event ticketing, including:

- Every ticket shall carry a globally unique identifier (UUID) and a scannable barcode or QR code for venue validation.
- Payment processing shall follow PCI-DSS standards, and refunds shall always route back to the original payment method.
- Attendee data such as names and emails shall be handled in line with applicable privacy regulations (GDPR, CCPA), including explicit opt-in for marketing communications.
- Digital ticket inventory shall never exceed fire code and building capacity limits for the specified venue.